

VAHDAM Lifecycle OS — Product Requirements Document

Status: v1.1 (complete) · **Owner:** Anchit Tandon (anchit.tandon@vahdam.com) · **Last updated:** 2026-07-06

Supersedes: v1.0 of 2026-07-02, and the v0.1 draft of 2026-06-04 (both preserved in git history — the draft documents the original consolidation thinking and is quoted where the "origin" of a feature matters).

What changed in v1.1 (2026-07-06): the V1/V2 version taxonomy (§1.4) is now recorded; the demo/mock access gate has been removed, so every signed-in user gets full live access (§9); the app UI is locked to a single forest-green theme, with the dark/dusk/light switcher removed (§7.1/§8); domain + OAuth migration tooling was added (§7.5); and the milestones (§11) run through 2026-07-06.

Live app: <https://vahdam-marketing-mailers-architect.vercel.app/> · **Presentation deck:** [/prd-deck](#) ·

Repo: [Anchit-AI-Hustle/vahdam-lifecycle-os](#)

0. Executive summary

VAHDAM Lifecycle OS is one application that runs the entire retention + acquisition growth loop for VAHDAM Teas: read the customer data → watch what competitors send → plan the month → generate brand-locked mailers, ads and landing pages → review, approve and learn — with an AI "Smart Brain" doing the daily loop automatically and a brand LLM ("ChaiGPT") that can operate every tool conversationally.

It replaced a scatter of disconnected tools (a dashboard here, a mailer builder there, competitor spreadsheets, ad-hoc briefs) with **one Vercel-hosted app, one login, one design system, one shared data layer** — built and operated at effectively **zero fixed infrastructure cost** (Vercel Hobby + Supabase free tier + free-tier LLM fallbacks), which is itself a deliberate engineering constraint that shaped the architecture (§7).

The system today: **11 user-facing modules, 12 serverless functions** (the exact Vercel Hobby cap), a **6-provider LLM waterfall**, a **4-provider image cascade**, **40+ database tables**, **376 active products across 3 regional catalogs**, an installable **PWA + native Android/iOS super-app shells** (code fully in this repo, auto-synced to production by architecture), and a daily **03:30 UTC autonomous planning cron** with human approval gates.

1. Vision

1.1 The one-line vision

*Read your data → see what competitors are doing → plan the calendar → write the email or ad → ship the landing page → let the Brain repeat it every day. **One app, one design system, one login***

— and eventually, one autopilot.

1.2 The final vision (north star)

The end state this project is building toward, in order of maturity:

1. **A self-driving lifecycle department.** The Smart Brain keeps a rolling 15-day campaign calendar per market, re-planned every morning against fresh data, festivals and competitor moves. Humans stop *producing* campaigns and start *approving* them: every slot arrives with the full funnel already generated — mailer, Meta/Google/TikTok ads, and a hosted landing page — and one click ships or rejects it. Rejection feedback trains the next plan.
2. **The brand's own AI employee.** ChaiGPT is the interface for everything: "plan a winback for lapsed UK tea drinkers and generate the assets" is a conversation, not a workflow. It operates the same tools the UI does, quotes real figures with evidence, and refuses to invent data.
3. **Closed-loop distribution.** Today generation stops at platform-ready objects (`push_status: not_integrated_phase_2`). Phase 2 wires Klaviyo (already fully scaffolded — every operation returns its exact would-be API request until a key is set) and the ad platforms so approved campaigns *send themselves*, and performance flows back into the Brain's winner-detection thresholds.
4. **Compounding intelligence.** Every asset generated, every competitor email captured, every approval/rejection and every campaign metric lands in the Knowledge Base — so the system's taste and hit-rate improve with use. The long-term moat is not any single generator; it is the accumulated, brand-locked corpus + feedback loop.
5. **Everywhere the team is.** The same OS installs as a PWA and ships as native Android/iOS super apps (one app containing the whole toolkit), so a plan can be approved from a phone.

1.3 Why "OS"

The name is deliberate. This is not a mailer tool with extras; it is an **operating system for the growth function**: shared auth/navigation shell, shared brand kernel (palette/typography/voice enforced in every generator), shared data layer, shared AI services — and applications (Analytics, Calendar, Studio, Competitor Intel, KB, Ads, Landing Pages, Brain, ChaiGPT) running on top of it.

1.4 Version taxonomy (V1 vs V2) — product-owner convention (2026-07-03)

As the OS matured, a two-generation convention was adopted so menus, hubs and this document can speak precisely about what is the legacy base versus the newer Lifecycle-OS layer:

- **V1 = the legacy base app** — everything that existed before 2026-07-03: the dashboard/analytics, the `/plan` 30-day RFM calendar, the Mailer Studio (`/studio`), Competitor Benchmarking, Knowledge Base, Ad Campaigns, Landing Pages, ChaiGPT, and the Smart Brain.
- **V2 = the Lifecycle OS additions of 2026-07-03** — the cohort mailer-calendar system (`/mailer-calendar`), the UK non-engagers campaign hub (`/uk-non-engagers`), tier-routed LLM/image cascades plus a video-core, the Social Media OS (`/social`), the retention/influencer knowledge library (`knowledge/retention/`), and the standing left-hand-nav information-architecture rule.

V1 features are upgraded by customising the base version only where needed. Where a capability exposes both generations, the earlier build is labelled **Draft 1** and the current one **Draft 2** (e.g. Plan Calendar V1 = Draft 1 vs Mailer Calendar V2 = Draft 2 of calendaring; Mailer Studio V1 = Draft 1 vs the

Mailer Calendar's built mailers = Draft 2 of mailer creation). The nav shows a quiet V1/V2 chip; Draft 1/2 lives in tooltips and info panels.

2. Origin story — how this project came to be

The chronology matters because every module was pulled into existence by a concrete operational pain, not speculatively:

1. **It started as the Mailer Studio** ([vahdam_mailer_architect_v34.html](#) — the "v34" records 34 iterations of a single-file app). The original need: producing brand-correct lifecycle emails was slow, inconsistent, and dependent on a designer's availability; generic AI output drifted off-brand (wrong greens, wrong fonts, "wellness journey" copy). The answer was a wizard that bakes the brand style guide into the generation itself.
2. **Generation needs targeting** → the **Data Analysis dashboard** was built so briefs come from RFM segments, cohort retention and send-time behavior instead of guesswork.
3. **Targeting needs cadence** → the **Marketing Calendar** turned analytics into a 30-day, festival-aware send plan that feeds the Studio one click per row.
4. **Cadence needs context** → the **Competitor Benchmarking** system began capturing every competitor email automatically (a dedicated Gmail inbox + IMAP sync into a Google Sheet), then grew ad libraries and landing-page tracking, because promo cadence decisions are made relative to the market.
5. **All of it needs memory** → the **Knowledge Base** became the single source of truth for the brand kit, historical winners, and reference assets that seed future generation.
6. **Email is half the funnel** → **Ad Campaigns** and **Landing Pages** extended the same brand-locked generation to paid acquisition and to the pages traffic lands on.
7. **The loop should run itself** → the **Smart Brain** (built as an MVP on 2026-06-10, hardened through June) wired all of the above into a persistent daily plan → generate → review loop with human approval.
8. **The whole stack should be conversational** → **ChaiGPT** (2026-06-26) put a tool-calling brand LLM on top of every core, and the **Klaviyo scaffold** prepared the send-side integration.
9. **It should be in your pocket** → PWA installability (2026-06-19) and native **Android/iOS Capacitor super apps** (2026-07-01).

Two structural decisions from the v0.1 draft era still define the product:

- **Consolidation over federation** (June): the separately-deployed competitor hub and sibling experiments were folded into this single repo/app, because "the team re-derives the same context in every tool" was the #1 pain.
 - **Stay within free limits until the value is proven**: Vercel Hobby's 12-function cap was treated as a design constraint, not a blocker — producing the [?action=](#) router architecture (§7.2) that still keeps marginal feature cost near zero.
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3. Problem statement

Before Lifecycle OS, VAHDAM's growth work was spread across disconnected surfaces. Concretely, the team:

- re-derived the same customer context in every tool (dashboard exports → slide decks → briefs → email builder);
- had no single place to see **their own plan next to competitor activity** (email *and* ads);
- produced creative that drifted off-brand — off-palette tints, wrong typefaces, banned discount-panic copy ("hurry", "last chance") — because every generator and freelancer started from scratch;
- had no shared, growing library of what was made and what worked;
- relied on fragile manual data syncs, and had **no always-on process**: if nobody planned the month, there was no plan.

The product bet: consolidate the loop into one app, enforce the brand at the generation layer (not at review time), and progressively automate the loop until humans only approve.

4. Personas & jobs-to-be-done

Persona	Jobs the OS does for them
Retention manager	Read RFM/cohort truth; approve the rolling calendar; trigger/inspect mailers; watch repeat-rate and winback performance.
Creative / copy	Generate brand-locked mailer + ad copy and imagery; browse the KB for anchors; clone and re-prompt past assets.
Growth / paid lead	Plan Google/Meta/TikTok campaigns and landing pages on one calendar; get baked-text ad creatives per size; benchmark competitor ads.
Competitive analyst	Browse every competitor mailer (full HTML + promo codes), ad-library deep links, tracked-brand registry, offers/funnel intel.
Founder / business leader	One URL (and one phone app) to see the whole growth function: data, plan, assets, competitor moves, and what the Brain proposes next.
The Smart Brain itself (machine persona)	First-class consumer of every API: it needs the same cores humans use, which is why all logic lives in shared modules rather than UI code.

5. The domains (Growth OS map)

The OS is organized into **eight domains**. Each domain has a UI surface, an API core, and (since June) a slash-command "team member" in `.claude/commands/` that operates it agentially:

#	Domain	Surface(s)	Backbone
1	Intelligence / Analytics	<code>/analytics</code> (Data Analysis), Cohort Definitions	client analytics engine + Supabase + offline DuckDB pipeline
2	Planning	<code>/plan</code> (30-day calendar), Smart Brain rolling 15-day plan	<code>calendar-generate</code> , <code>smart-brain-plan</code> , festivals data
3	Creative — Email	<code>/studio</code> (Mailer Studio)	5-stage AI pipeline, 11 archetypes, 4 variants
4	Creative — Paid	<code>/ads</code> (Ad Campaigns), <code>/landing</code> (Landing Pages)	ad compositor + LP compiler + <code>/lp/:id</code> hosting
5	Competitive Intelligence	<code>/competitor</code> (Benchmarking hub)	IMAP capture → Sheet/Supabase <code>ci_*</code> , ad libraries, workers
6	Knowledge / Memory	<code>/kb</code> (Knowledge Base), <code>/assets</code> (Created Assets)	<code>kb_*</code> tables, ingest-with-AI, asset hub
7	Autonomy	<code>/brain</code> (Smart Brain console), daily cron	6 services + rolling plan + HITL review
8	Conversation / Voice	<code>/chaigpt</code> (ChaiGPT), <code>/agent</code> (Vahdam Agent)	brand-llm tool loop, evidence contract, TTS

Cross-cutting: **Brand kernel** (§8), **Klaviyo lifecycle** (scaffolded, §6.12), **Mobile super apps** (§6.14), **Ship/Deploy** (Vercel).

6. Feature specifications

Every feature below follows the same lens: **Origin** → **Need** → **Purpose** → **What it does** → **How it works** → **Future**.

6.1 Home — the portfolio hub (`/`, `index.html`)

- **Origin.** As modules multiplied, the home page was rebuilt (2026-07-01) from a static landing page into an "interactive portfolio hub" after the team kept losing track of what existed.
- **Need.** Eleven tools are only useful if you can find them; new stakeholders need a 30-second mental model.
- **Purpose.** One screen that always mirrors the entire OS and teaches the pipeline.
- **What it does.** Card grid of every tool with **Open / Peek** (live iframe preview) actions, section filter chips (Agents & Data / Knowledge Base / Create), a **⌘K command palette** ("Jump to any tool..."), a 4-step pipeline strip (*Look at data* → *Plan a month* → *Write emails* → *Send & compare*) with a **Run Autopilot** shortcut into the Smart Brain, and a live health dot polling `/api/public-config?health=1`.
- **How it works.** Pure client JS; the card grid is generated from `window.__LC_NAV` — the same nav model `auth.js` renders — so the homepage can never drift out of sync with the sidebar.
- **Future.** Role-based default views; surfacing "what the Brain did overnight" as the hero.

6.2 Data Analysis (/analytics , dashboard.html)

- **Origin.** Built immediately after the Mailer Studio because generation without targeting produced pretty emails to the wrong people.
- **Need.** Retention decisions (who to mail, when, with what) were made from stale exports; nobody could answer "which segment is drifting?" live.
- **Purpose.** The intelligence layer that drives every plan and brief downstream — "see who buys, who drifts, what to send next."
- **What it does.** Seven analysis views: **Revenue Pulse** (KPIs, daily trend, region mix), **Campaigns Deep-Dive** (email/SMS subject & content performance), **Audience Segments (9 RFM segments + behavioral overlays: discount sensitivity, category affinity, engagement decay 30-vs-90-day)**, **Product Intelligence** (promoted-vs-bought, cross-sell map), **Send-Time & Behavior** (heatmaps, purchase windows, seasonality), **Cohorts & Retention** (triangle, curves, lifecycle funnel), and **Strategic Insights** (auto-generated, ranked by impact). Region (US/UK/Global/India) and currency (USD/INR) switches; trend granularity day → year.
- **Data in/out.** Three source modes — synthetic seed (demo), **CSV/XLSX upload** (append or replace), or **Link a database** (Supabase/REST endpoint; analysis then scopes to that DB only until reset). Exports CSV and color-preserving XLSX. Results persist to localStorage and feed the Calendar and Studio.
- **How it works.** Fully client-side analytics engine (~2,300 lines) with ApexCharts; a reproducible Google-Sheets dashboard generator (2026-06-30) mirrors the key views into a shareable Sheet. The offline **DTC data engine** (`ingest/` , DuckDB → `dtc.*` Supabase tables) supplies real Matrixify/Shopify/Klaviyo/WebEngage exports.
- **Future.** Warehouse connectors via server proxy; scheduled refresh; anomaly alerts pushed into the Brain's daily insights.

6.3 Cohort Definitions (/cohorts , cohort-definitions.html)

- **Origin.** Written down after segment names started meaning different things to different people ("lapsed" = 60 or 120 days?).
- **Need.** A canonical, human-readable segmentation contract shared by analytics, calendar and generation.
- **What it does.** Documents **12 lifecycle cohorts across 5 stages** — NEW (Prospect, First-Time Buyer 0–45d, Second-Order Pending), ACTIVE (Repeat/Habitual, Growing Basket, Subscriber), VIP (Loyal Core, High-Value VIP), RISK (At-Risk Regular, One-and-Done Stalling), LAPSED (Lapsed Buyer, Dormant/Churned) — each with definition, audience mindset, and trigger strategy. The Brain's runtime engine uses a compatible operational set (Champions, Loyalists, New Buyers, Winback High-LTV, At-Risk, Nurture).
- **Future.** Make definitions executable (one JSON contract consumed by dashboard + Brain instead of parallel definitions).

6.4 Marketing Calendar (/plan , calendar.html)

- **Origin.** The step that connected analytics to production: once segments were visible, the question became "so what goes out on Tuesday?"

- **Need.** Month planning lived in spreadsheets, disconnected from both the data and the email builder; festivals were remembered too late.
- **Purpose.** Turn the analytics state into a concrete, capacity-constrained 30-day send plan — then hand each row to the Studio with one click.
- **What it does.** Generates a **30-day plan** across selected markets (US/UK/Global/IN) with per-market weekly send capacity: each entry = date × market × segment (with live segment size) × content type (promo/editorial/lifecycle/launch/winback) × one of the 11 layout archetypes × hero product × subject-line hint, festival-weighted (`data/festivals.json`) and send-time tuned. Renders a month grid + filterable table; CSV export; a compiled mini-dashboard (revenue, open rate, active customers, top segment) on top; **"Generate mailer"** pushes any row through the full strategy → variants pipeline.
- **How it works.** `POST /api/calendar?action=generate` (`_shared/calendar-generate.js`) consumes the analytics summary from localStorage; `?action=trigger-mailer` (`calendar-trigger.js`) feeds the pipeline of §6.5.
- **Future.** Fully absorbed by the Smart Brain's rolling plan (§6.9) — the 30-day generator remains the manual/what-if mode and the 5-scenario engine's base.

6.5 Mailer Studio (`/studio` , `vahdam_mailer_architect_v34.html`)

- **Origin.** The founding application — 34 versions of iteration compressed into its filename. Every other module exists to feed or learn from it.
- **Need.** Brand-correct lifecycle emails took days and still drifted: wrong palette tints, wrong fonts, banned urgency copy, inconsistent structure. Generic AI tools made it worse — fluent but off-brand.
- **Purpose.** Produce **production-ready, brand-locked email HTML in minutes**, with enough structural variety that A/B tests are real tests.
- **What it does.** A **5-step wizard**: Brief (free text or "Create Brief with AI" → a director-grade 450–600-word brief; multi-market targeting; 8 campaign types) → Products (auto-picked from the real regional catalog or manual) → Generation → Review & Refine → Final HTML. Produces **4 variants per campaign**: **A** (Image · hero close-up, conversion-led), **B** (Image · lifestyle wide, story-led — structural divergence from A is *forced*, not suggested), **T1** (Text · editorial), **T2** (Text · founder note). **11 layout archetypes** (hero-led-editorial, product-grid-conversion, storytelling-narrative, single-product-spotlight, gift-bundle-showcase, ritual-journey, comparison-discovery, founder-note, editorial-trend-roundup, limited-drop-countdown, subscription-anchor) are mapped per campaign-type × variant. Output is compact (~1200–1500px, "two scrolls"), passes a client-side `brandPaletteCheck()` , uses region-aware store URLs/currency, and scores itself before display. A Budget/Max-Power **tier dial** and an in-page assistant composer round it out.
- **How it works.** Two paths share one contract: a client-side archetype renderer, and the server **5-stage pipeline** (`api/ai/pipeline/`): **strategy** (Master Strategic Lock: audience truth, product picks, strategy type, image-style lock, A/B concepts — think) → **variant** ×2 (lock*: pure execution, mandatory 7-section structure for A, structural opposite for B) → **images** (up to 3: hero/product/lifestyle, with retries and cascade §7.4) → **html** (responsive email HTML; offer above fold; price always visible; max 7 sections) → **score** (strategy alignment / content density / copy quality / variant divergence; pass gate ≥7, divergence ≥8; failing variants signal a retry). The "think → lock → execute" order exists because earlier "generate then patch" attempts produced incoherent emails.
- **Claims.** Brief-to-4-variants in minutes; zero off-palette output by construction; every mailer traceable to a strategy JSON.

- **Future.** KB-driven few-shot conditioning (retrieval of past winners per archetype/market); Klaviyo template push (Phase 2).

6.6 Competitor Benchmarking & Competitive Intelligence (`/competitor`, `competitor-benchmarking.html`)

- **Origin.** Started as a separately-deployed "competitor hub" reading a Gmail inbox; merged into the OS during consolidation, then expanded into a full CI system (June 17–20) with off-Vercel collectors.
- **Need.** Promo-cadence and creative decisions are relative: "should we discount this week?" depends on what Palais des Thés, Art of Tea and 30+ others are sending *right now*. Manually forwarding competitor emails didn't scale and lost the HTML.
- **Purpose.** An always-fresh, searchable archive of competitor lifecycle activity — mailers, ads, landing pages, offers — that both humans and the Brain benchmark against (informing prioritization but *never* qualifying our own winners — the two streams are deliberately isolated).
- **What it does.** Tabs: **Discover Brands** (registry of tracked brands across Tea/Coffee/Supplements × US/UK; live seeded list), **Mailers** (every captured email: brand, subject, promo codes, received-at, full stored HTML rendered in-place, attachments/screenshots; search + platform/promo filters; live 45s polling), **Meta / Google / TikTok Ads** (deep links per tracked brand into the free public ad libraries; collected ads via workers), **Landing Pages, Insights**. Also serves the Mailer Discovery view (`/discover`).
- **How it works.** A dedicated capture inbox is auto-subscribed to competitor lists by a local **Playwright worker** (`workers/auto-subscribe.js`); Gmail **IMAP** (`imapflow` + `mailparser`) ingestion parses each mail (subject, body, promo-code regex, inline images, attachments) into the Google-Sheet database (columns A–K, raw HTML capped at 49k) and mirrors into Supabase `ci_emails`; `?action=poll` gives near-real-time freshness while the page is open; a Cloudflare-Email-Routing webhook (`?action=ingest`) offers a zero-cost push path; `ci-collect-*` actions + off-Vercel collectors (`collect-ads.js`, `collect-landing.js`, `collect-wayback.js`) populate ads/landing/offers; `ci-enrich` runs lazy LLM tagging; `ci-funnel` reconstructs funnels. **Auth to Google is keyless** via Workload Identity Federation (\$7.5).
- **Claims.** Competitor emails appear in the hub ~minutes after they land; the archive keeps full render-quality HTML, not screenshots of screenshots; \$0/month capture stack.
- **Future.** Scale to 50+ sources with a queue+worker model; move artifacts to object storage + CDN; automated creative-tagging into the KB schema.

6.7 Knowledge Base (`/kb`, `knowledge-base.html`)

- **Origin.** Grew out of a June restructure ("collapse to VAHDAM + Competitors") when reference material was scattered across drives and chats.
- **Need.** Generators are only as good as their references; the brand needed one growing library of its own assets and curated winners that generation can be anchored to.
- **Purpose.** The **single source of truth every generator reads from** — brand kit, markets, product catalog, historical top emails with metrics, uploaded references, ingested URLs.
- **What it does.** Channel tabs (Mailers / Meta / Google / TikTok / Landing Pages) × region filters; **file/folder upload**; **"Ingest URL with AI"** (fetch → strip → LLM summary with key points + tags); an import path for historical **top emails with performance metrics** (they become style + voice anchors);

tracked-brand management; email-type classification (text/html/image-heavy) over the captured competitor corpus.

- **How it works.** `api/kb.js` router (`ingest|list|top-emails|brands|classify-emails`) over Supabase (`kb_knowledge`, `kb_files`, `kb_top_emails`, `competitor_brands`, `competitor_emails_classified`).
- **Future.** Vector embeddings + "find designs like this" retrieval; RAG-conditioning of the Studio and ad builder on brand-compliant exemplars (learn patterns, never copy).

6.8 Ad Campaigns (`/ads`, `ad-campaigns.html`) & Landing Pages (`/landing`, `landing-pages.html`)

- **Origin.** Email covered retention; paid acquisition briefs still lived in slides. Both modules extend the same brand kernel to paid. The landing-page system additionally absorbed a real campaign need — the Ashwagandha-coffee presell funnel — which produced the `/lp/*` hosted-page contract and the battle-tested LP compiler.
- **Need.** Ad creative was produced ad-hoc per platform and size; landing pages required a developer for every campaign; neither enforced the brand system; and **Aman's P01 mandate** — "*sell happiness, not ingredients*" (target women 45+/busy mums, emotional end-state, 1-second scroll-stop, baked-in offer text) — needed to be encoded, not remembered.
- **Purpose.** One calendar for paid; per-platform ad objects with **copy baked into the creative**; landing pages generated in the same brand system as the email/ad that drives traffic to them.
- **What it does.**
- **Ads:** compiled KPI dashboard (spend, ROAS, CPA, top channel); tabs Calendar / Google / Meta / TikTok; a one-prompt **autofill** that populates every field; a client-side compositor that renders **one still PNG per ad size with the text overlay baked in** (Google 1200×628 & 1200×1200; Meta/IG 1080×1080 & 1080×1920; TikTok 1080×1920). Shipped "sell-happiness" reference creatives (Meta static + Reels video) live in the repo.
- **Landing Pages:** brief-to-page generation per channel × region, linked back to the mailer so one brief feeds both surfaces; a library of saved pages with **clone-into-editable-entry**; generated pages are **hosted immediately at `/lp/:campaignId`** (dynamic serving via `api/calendar?action=lp`, downloadable as a self-contained file); flagship hand-tuned funnels at `/lp/cortisol-v1|v2`, `/lp/agent`, `/lp/best` — including the all-in-one **voice-agent landing page** where an embedded agent narrates and asks the visitor questions.
- **How it works.** `api/ai/generate.js` (copy) + `api/ai/image.js mode:'ad'` (creatives) + `_shared/lp-compiler.js` (themes × funnel variants, ported from the standalone `marketing_automation/` React compiler); persistence in `ads_generated` / `landing_pages_generated`.
- **Future.** Phase-2 platform push (Meta/Google APIs); LP performance capture feeding the Brain.

6.9 Smart Brain — the autonomous daily loop (`/brain`, `smart-brain.html`)

- **Origin.** Built as an MVP on **2026-06-10** ("Build Smart Brain MVP") once every manual step existed end-to-end; hardened through June (persistent plan → funnel preview → creatives → dual-mode → scenarios).

- **Need.** The loop worked but only when a human ran it. Growth cadence shouldn't depend on someone remembering to plan; and the "plan" needed to survive across days, not be regenerated from scratch (losing human edits).
- **Purpose. The autopilot:** keep a persistent, rolling **15-day calendar per market**, re-reviewed every morning against fresh data, with humans approving — and every approval generating the *complete funnel* automatically.
- **What it does.** Console shows mode (DB-linked/preview), plan entries, pending approvals, last sync; the rolling calendar table (date, market, cohort + live size, objective + hero product, channels, **confidence score**, status pill); **View** previews the full funnel *before* approval (mailer iframe, landing page, per-platform ads with AI creative images, JSON details); **Approve** LLM-writes mailer + Meta/Google/TikTok ads + landing page (hosted at `/lp/:id`), persists to `smart_generated_campaigns` and mirrors to `ads_generated/landing_pages_generated`; **Reject** (with notes) feeds recalibration. **Daily insights** and **changes-from-last-sync** panels narrate what the Brain changed and why. **Dual-Mode Generation** panel: Budget vs Max-Power model tier, an 8-stage traced **agentic run** (data → analysis → planning → calendar → content → assets → review → ideation), and a **5-scenario calendar** (best/medium/conservative/emergency/instant pre-staged; a standby scenario can be activated in one click).
- **How it works.** `lib/smart-brain/services.js` — **6 services**: KnowledgeBase (index own campaigns/assets/metrics), Analysis (RFM cohorts; winner detection vs explicit thresholds, e.g. email open ≥22%, click ≥1.8%; product scoring; channel benchmarks), CompetitorBenchmarking (isolated stream — informs prioritization, never qualifies own winners), CalendarIntelligence (festival-aware 15-day plan with confidence + MVT plan; every entry `needs_human_verification`), Generation (audience/retargeting/lookalike specs, funnel, email+LP+ads as platform-ready objects, `push_status:'not_integrated_phase_2'`; `safeCopy` scrubs the banned-phrase list), Review (daily automated review + **mandatory weekly human recalibration**). `_shared/smart-brain-plan.js` makes the plan *persistent*: daily sync **diff-updates only tentative / rejected entries** — a human approval mid-window always wins over the machine. Daily **Vercel Cron at 03:30 UTC** hits `/api/brain?action=cron` (`CRON_SECRET`-protected). AI creative images are hosted on Supabase Storage.
- **Claims.** A tentative, evidence-scored 15-day plan exists every morning without human input; approval-to-full-funnel is one click; nothing sends without a human (by design, Phase-2 gate).
- **Future.** Phase-2 push to Klaviyo/ad platforms; auto-approve above confidence thresholds with spend caps; performance-informed replanning.

6.10 ChaiGPT — the brand LLM (`/chaigpt`, `chaigpt.html`)

- **Origin.** 2026-06-26: "brand LLM that operates the whole growth stack." The insight: the OS had become an API surface an LLM could *drive* — so give the brand its own Claude.
- **Need.** Every tool answers one question; leadership questions span tools ("what's our best cohort, what are competitors doing about it, and what should we send?"). Also: answers must be *grounded* — no invented figures.
- **Purpose.** A conversational operator for the entire stack, with evidence discipline.
- **What it does.** Claude-style chat (market selector, suggestion chips, Klaviyo status pill) that plans multi-step work, calls real tools, and shows a collapsible **tool trace** (every call + args + results). **11 registered tools**: `ask_analytics`, `run_analysis`, `list_cohorts`, `get_calendar`,

`get_competitor_benchmarks`, `search_knowledge_base`, `list_campaigns` (reads) + `generate_calendar`, `generate_assets_for_slot`, `run_agentic_campaign`, `klaviyo` (writes — only on explicit ask). The system prompt enforces a **4-point evidence contract**: quote exact tool-sourced figures, name the target metric + expected impact, state a complete hypothesis, quote competitor benchmarks.

- **How it works.** `_shared/brand-llm.js`: a provider-agnostic **tool-calling loop** — the model emits strict JSON (`{action:'tool'|'tools' [≤3 in parallel]|'final'}`), the server executes against the same `_shared` cores the UI routes use, feeds results back, loops (default 5 steps). Because tool calls are plain JSON rather than a provider function-calling API, **the loop works across the entire 6-provider waterfall including free tiers**. Speed: pins the first provider that answers, dedupes repeated tool+args, 20s per-provider timeout. The product name is one constant (`BRAND_LLM_NAME`) — rebrandable in one line.
- **Future.** Memory across sessions; scheduled "morning briefing" messages; Klaviyo write-ops going live with the key.

6.11 Vahdam Agent — buyer-facing voice + chat (`/agent`, `agent.html`, `agent-widget.js`)

- **Origin.** June 13–21: voice/chat agents grew from the landing-page work (an LP that *talks* converts differently) into an app-wide, embeddable concierge.
- **Need.** Distinct from ChaiGPT (internal operator): buyers need a persuasive, KB-grounded concierge that behaves like a telecaller and never leaks internal data.
- **What it does.** Full-screen voice+chat agent; per-type agent creation (brand/collection/product/persona); female-voice default TTS; lives on the agent-integrated landing page and embeds on Shopify with one script tag; spoken-friendly, no-markdown replies. Guardrails: evidence policy + confidentiality firewall (buyer-safe vs internal data classification matrix), anti-scraping, persuasive persona playbook.
- **How it works.** `brain-agent.js` (buyer `agent-chat` vs internal `team-chat` routes), `evidence-policy.js`, `data-classification.js`; TTS via `?action=tts` (ElevenLabs proxy with browser-voice fallback).
- **Future.** Order-aware conversations once commerce APIs are connected; multilingual voice.

6.12 Klaviyo integration (scaffolded)

- **Origin.** Built alongside ChaiGPT (2026-06-26) so the send-side integration would be *ready* rather than hypothetical.
- **Need.** Klaviyo is the ESP; Phase 2 (closed-loop sending) needs profiles/segments/campaigns/flows/reporting access.
- **What it does.** `_shared/klaviyo-core.js` mirrors Klaviyo's JSON:API (revision 2024-10-15): **15 operations** — status, profiles, lists, segments, metrics, events, campaigns, flows, templates, campaign reporting, create-profile, subscribe, track-event. **Until `KLAVIYO_API_KEY` is set, every op returns `{connected:false, would_request:{method,url,body}}`** — a structured dry-run stub — so ChaiGPT, the tools UI and the chat flows all work end-to-end today and go live by adding one env var.
- **Claims.** Zero-risk integration: the full request surface is inspectable before any key exists.

6.13 Created Assets hub (/assets , assets.html)

- **Need.** Generated output (mailers, ads, LPs) was scattered per-tool; reuse meant regenerating.
- **What it does.** One searchable, filterable library of **every asset the suite has produced** — preview, download HTML, **clone**, and **copy a portable AI prompt** for any asset (so it can be re-run in any external LLM), categorized by type/market; local-first with Supabase sync.

6.14 Mobile super apps — PWA + native Android & iOS

- **Origin.** PWA installability shipped 2026-06-19; native **Capacitor** apps ("Add iOS + Android native apps via Capacitor — WebView shell over live PWA") shipped 2026-07-01.
- **Need.** Approvals and monitoring shouldn't require a laptop; leadership wants the whole OS as **one app on the phone** (a true super app: all eleven tools in one shell).
- **Purpose.** The entire Lifecycle OS, installable from a store or a link, always up-to-date.
- **What it does & how it works.**
- **PWA:** `manifest.webmanifest` + `sw.js` (`lifecycle-os-v16`) make the web app installable on Android/iOS with an offline shell; navigation is network-first with cache fallback; `/api/*` is never cached; the SW self-heals stale caches.
- **Native shells:** `capacitor.config.json` (appld `com.vahdam.lifecycleos` , appName "VAHDAM Lifecycle OS") with `server.url` **pointed at the production deployment** — the app is a hardened WebView over the live site. **This is the auto-sync guarantee:** every web deploy *is* a mobile release; the binaries never go stale because the code they render is served from production. **The complete native project code lives in this repo** — `android/` (Gradle project, minSdk 22 / target 34), `ios/` (Xcode project + Podfile), `mobile/` (self-contained Capacitor sub-project), `mobile-shell/` (web dir) — no separate mobile repo to keep in sync.
- **Downloadable builds:** the **Mobile Builds** GitHub Actions workflow (`.github/workflows/mobile-builds.yml`) builds the binaries and publishes them to the fixed release tag `mobile-latest` , giving stable download links:
- **Android APK (installable directly):** <https://github.com/Anchit-AI-Hustle/vahdam-lifecycle-os/releases/download/mobile-latest/vahdam-lifecycle-os.apk>
- **iOS app (unsigned IPA — sideload via AltStore/Sideloadly, or sign for TestFlight):** <https://github.com/Anchit-AI-Hustle/vahdam-lifecycle-os/releases/download/mobile-latest/vahdam-lifecycle-os-ios-unsigned.ipa>
- **All builds page:** <https://github.com/Anchit-AI-Hustle/vahdam-lifecycle-os/releases/tag/mobile-latest>
- Run the workflow once from the **Actions** → **Mobile Builds** tab (or push a change under `android/` / `ios/`) to mint/refresh the binaries; it re-publishes to the same links every time.
- **Future.** Play Store / App Store listings (store metadata already scaffolded); push notifications for pending approvals; iOS signing pipeline.

6.15 Growth OS slash-command team (`.claude/commands/`)

- **Origin.** 2026-06-21 ("Integrate growth-team slash commands + connector/skill routing") — the realization that the OS's cores + hosted connectors (Klaviyo, Canva, Figma, Ahrefs, Higgsfield...) compose into an agentic *team*.

- **What it does.** Project slash commands operate the brand as a full growth team: `/growth-team` (router), `/campaign-plan`, `/email-flow`, `/mailer`, `/ad-creative`, `/landing-page`, `/design`, `/shopify` (public storefront scraping — Admin API deliberately not authorized), `/analytics`, `/competitor`, `/seo`, `/db`, `/ship`. **Every command enforces the Brand Constants (§8) and the P01 mandate.**
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7. Platform architecture

7.1 Shape

- **Frontend:** independent, self-contained static HTML pages (inline CSS/JS; the Studio alone is ~7,700 lines) sharing one auth/nav shell (`auth.js`) and localStorage state. No framework, no build step for pages — deliberate: any page can be understood, patched and shipped in isolation. The UI is locked to a **single forest-green theme** (`theme.css`): the earlier dark/dusk/light switcher was removed on 2026-07-06, so every page renders on the brand's forest-green canvas with gold/cream text and cannot drift to an off-brand light mode.
- **Backend:** Vercel serverless functions under `api/`; heavy logic in `api/_shared/` (underscore paths don't count against the function cap) and `lib/`.
- **Deploy:** single Vercel project (`framework: null`); `npm run build` only rebuilds product catalogs; friendly URLs via `vercel.json` rewrites; CI = HTML smoke check + syntax/function-cap/Playwright gates.

7.2 The 12-function constraint (and why the architecture looks like this)

Vercel Hobby caps a project at **12 serverless functions**. The app sits exactly at the cap: `ai/generate`, `ai/image`, `ai/pipeline/{strategy,variant,images,html,score}`, `brain`, `calendar`, `competitor`, `kb`, `public-config`. This forced three patterns that are now product strengths:

1. **?action= catch-all routers** — `api/brain.js` alone dispatches ~30 actions; `competitor.js` and `kb.js` likewise. New capabilities cost zero functions.
2. **_shared/ modules** — all real logic is import-cost-free and reusable by every router *and* by ChaiGPT's tools (one implementation, three consumers: UI, API, LLM).
3. **Rewrite-collapsing** — retired endpoints (`/api/health`, `/api/voice`, `/api/klaviyo`, `/api/smart-brain`) live on as rewrites into surviving routers, so URLs never break.

7.3 The 6-provider LLM waterfall (`_shared/llm.js`)

All text generation goes through one caller: **OpenAI** (3 rotating keys) → **Anthropic** → **Gemini** (free) → **Grok/xAI** → **Groq** (free) → **Cerebras** (free). Features: per-call provider pinning (ChaiGPT's speed trick), Budget/Max-Power tier forcing, quota detection on **HTTP 400 + billing keywords** (both OpenAI and Anthropic return 400, not 429, on exhausted credit — a hard-won lesson), BOM/zero-width stripping, multi-strategy JSON parsing. **Consequence:** the platform *degrades to free tiers instead of failing*, and has no single-vendor dependency.

7.4 The image cascade (`api/ai/image.js`)

Gemini native (image-out models) → **Gemini Imagen** (paid tier only) → **OpenAI gpt-image-1** (3 keys) → **Pollinations** (free flux-pro → flux-realism → flux) → **on-brand placeholder data-URI** (callers never receive a broken tile). Three prompt modes (photo / full email design / ad-with-baked-text); region-aware currency; per-platform ad sizes.

7.5 Keyless Google auth — Workload Identity Federation

The competitor Sheet is accessed **without any stored Google key**: Vercel mints a per-request OIDC token → Google STS exchanges it → the code impersonates the service account (1h cached). Legacy JSON-key mode remains as fallback. Shipped 2026-06-07; documented in `docs/workload-identity-federation.md`.

7.6 Persistence

- **Supabase (Postgres)**: auth, KB, CI, Smart Brain (~25 timestamped migrations; 40+ tables incl. 14 `smart_`, 9 `ci_`, `kb_`, `ads_generated`, `landing_pages_generated`, `dtc.` facts; `COMBINED_RUN_THIS.sql` apply-all bundle; RLS hardened; service-role keys never reach the client — `/api/public-config` exposes only URL + anon key).
- **Google Sheet**: the legacy competitor-email database (columns A–K contract).
- **localStorage**: analytics → calendar → studio state handoff; asset drafts.
- **Supabase Storage**: AI creative images for generated campaigns.

7.7 Offline data engines (local, not on Vercel)

`ingest/` (Matrixify/Shopify/Klaviyo/WebEngage → DuckDB → Supabase `dtc.*`), `mailer_system/` (Python Claude-API campaign engine with thresholds), `marketing_automation/` (React+Express campaign compiler — origin of the LP compiler), `workers/` (Playwright collectors that hold no cloud keys and POST back to the app).

7.8 Product catalogs

Built at every deploy from Shopify exports: **US 173 · UK 101 · Global 102 active products** → compact JSON with handle, price, compare-at, tasting notes, caffeine, cups, category tags; served with CORS + long cache. Verified store URLs per market (US/UK/IN/EU/AU/Global) drive region-correct PDP/collection links in every generated asset.

8. The brand kernel (non-negotiables enforced in code)

Source of truth: `Brand style guide.pdf`, codified in `_shared/master-prompt.js` and enforced client-side (`brandPaletteCheck`), server-side (`safeCopy` scrub), and in every prompt:

- **Palette — ONLY four colors**: `#004A2B` forest green · `#AB8743` gold · `#171717` near-black · `#FBF5EA` cream. Known drift tints are explicitly banned and were purged.
- **Typography**: headings **Lao MN**, body **Proxima Nova** (with exact fallback stacks and `@font-face` sources). No substitute primaries.
- **Banned phrases**: "wellness journey", "transform", "liquid gold", "game-changer", "LIMITED TIME" (caps), "hurry", "don't miss out", "last chance", "while supplies last".

- **Preferred vocabulary:** ritual, restore, balance, origin, single-estate, hand-picked, steep, heritage, crafted.
- **Voice:** warm, sensory, story-driven; testimonials read as tiny personal stories with region-matched names.
- **P01 ad mandate (Aman):** *sell happiness, not ingredients* — emotional end-state creative for women 45+/busy mums, 1-second scroll-stop, offer text baked into the image ("Starter Pack 65% OFF + free frother + scoop").

Claim: brand compliance is a *property of the system*, not a review step. An asset that violates the palette or copy rules is corrected or rejected before a human ever sees it.

9. Security, auth & data governance

- **App auth:** Google OAuth (PKCE) via Supabase on the shared shell; sessions persist; the login wall is currently relaxed for the internal team (Mailer Studio always open by design); per-page gating can be re-enabled by flag. **Access mode (updated 2026-07-06):** the former demo/mock gate — which put non-@vahdam.com signed-in accounts into a simulated read-only mode (a `window.fetch` guard that faked write/generation responses, plus a demo banner) — was **removed**. Every signed-in user now gets full, live access; the `mockMode` / `__VAHDAM MOCK__` flags remain pinned to `false` for any external reader.
 - **Domain + OAuth migration:** moving a project to its `<slug>.anchit-tandon.com` domain is handled by `scripts/migrate-domains`. (*Vercel domain + GoDaddy CNAME*) which then hands off to `scripts/migrate-oauth`. Because Google sign-in is Supabase-mediated, the change that actually matters is the Supabase Auth redirect allowlist (auto-applied via the Supabase Management API); the Google web-client's JavaScript origin is a Console-only step (no gcloud/API can edit a web OAuth client), for which the tooling emits an exact plan. Detail in `docs/oauth-redirect-migration.md`.
 - **Secrets:** live only in Vercel env vars; `/api/public-config` never exposes service-role keys; provider keys never reach the browser.
 - **Cron protection:** `CRON_SECRET` guards `?action=cron` and sync; optional `INGEST_TOKEN` guards the email webhook.
 - **Keyless where possible:** WIF (\$7.5) removed the standing Google key.
 - **Buyer/internal firewall:** the buyer-facing agent runs behind a data-classification matrix (buyer-safe ↔ internal) + evidence policy; internal metrics can't leak into customer chat.
 - **Service worker:** never caches `/api/*`; HTML served `must-revalidate`.
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10. Success metrics

Product/ops metrics

- Time-to-first-insight; brief-to-4-variants time (target: minutes); approval-to-full-funnel time (one click + generation).

- % of calendar slots planned by the Brain vs manually; approval rate of Brain proposals; rejection-feedback incorporation.
- Competitor coverage: brands tracked, emails/ads captured per week, capture freshness.
- Brand compliance: off-palette/banned-phrase incidents reaching human review (target: 0, by construction).
- Platform: function count ≤ 12 ; \$0 fixed infra; provider-failure resilience (generation success rate despite quota exhaustion).

Business metrics (once Phase 2 closes the loop)

- Repeat rate, winback conversion, campaign-attributed revenue share, AOV expansion per cohort; email open $\geq 22\%$ / click $\geq 1.8\%$ thresholds already encoded as the Brain's winner gates.

11. Milestones (from the commit record)

Date (2026)	Milestone
~May 30	Per-project GCP/Gemini key infrastructure across the app portfolio; Mailer Studio v34 exists as the founding app
Jun 07	Lifecycle OS repo takes shape: WIF keyless Google auth; KB restructure (VAHDAM + Competitors); competitor capture hardening; one-prompt autofill across ads + LPs
Jun 09	Mailer Studio consolidated to one file at <code>/studio</code> ; competitor auto-subscribe Playwright worker
Jun 10	Smart Brain MVP + persistent 15-day rolling plan, daily cron, approve → generate, hosted LPs
Jun 13	Voice/chat agents; final presell landing pages with embedded voice agent (<code>/lp/*</code>)
Jun 17–18	Competitive Intelligence system (Supabase <code>ci_*</code> , off-Vercel collectors, Sheet → CI bridge); app-wide KB-grounded voice agent
Jun 19	Funnel preview <i>before</i> approval; AI creative + copy for every generated asset; PWA installable ; Capacitor scaffold
Jun 20–21	Smart Brain hardening + master-prompt foundation; brand asset code engine enforced across all generators ; P01 "sell happiness" mandate baked into ad creatives; growth-team slash commands ; first PRD deck; dual-mode Budget/Max-Power
Jun 22–25	Scenario-calendar engine (5 scenarios); consolidation merges; CI quality gates (syntax, function-cap, Playwright); strict palette lock; agent LLM-first
Jun 26	ChaiGPT — the brand LLM operating the whole stack + Klaviyo scaffold
Jun 27–30	LP cloning; reproducible Google-Sheets dashboard generator
Jul 01	Native Android + iOS super apps (Capacitor over live PWA); interactive portfolio-hub homepage; ChaiGPT faster loop + evidence answers; asset hub with portable prompts
Jul 02	This PRD v1.0 + business presentation deck ; Mobile Builds workflow with downloadable APK/IPA links
Jul 03	V2 Lifecycle OS layer (§1.4): cohort Mailer Calendar (<code>/mailer-calendar</code>), UK Non-Engagers Hub (<code>/uk-non-engagers</code>), tier-routed LLM/image cascades + video-core, Social Media OS (<code>/social</code>), retention/influencer knowledge library, and the LHS-nav IA rule; V1/V2 taxonomy adopted
Jul 06	Demo/mock access gate removed (every signed-in user gets live access); UI locked to a single forest-green theme (dark/dusk/light switcher removed); domain + OAuth migration tooling (<code>migrate-domains</code> / <code>migrate-oauth</code>); brand-compliant campaign mailer pack ; PRD v1.1 + refreshed decks (PPT/PDF/Word)

~190 commits, ~27 days of build velocity, 60+ merged PRs.

12. Roadmap

Phase 0 — Consolidation  *done*: one app, one nav, one design system; competitor hub merged; near-real-time capture; KB; brand kernel enforced.

Phase 1 — Autonomy with human gates  *done*: Smart Brain rolling plan + daily cron + full-funnel generation + HITL approve/reject; ChaiGPT operator; dual-mode tiers; scenario calendars; mobile super apps.

Phase 2 — Closed loop (next)

- Klaviyo live (add the key; the scaffold already dry-runs every request) — push approved mailers as campaigns/templates; sync segments both ways.
- Ad-platform export (Meta/Google) where APIs allow; LP performance capture.
- Store listings for the Android/iOS apps; push notifications for pending approvals.
- Auto-approve above confidence thresholds with budget caps (opt-in, per-market).

Phase 3 — Compounding intelligence

- KB embeddings + RAG conditioning of all generators on brand-compliant exemplars.
- CI at 50+ sources (queue+worker), object storage + CDN for artifacts, observability/alerting.
- Performance-informed replanning (the Brain reads what actually happened, not just what shipped).

Phase 4 — Scale-out

- Multi-brand/white-label (the brand kernel is already one module; ChaiGPT is renameable via one constant).
- Warehouse connectors; role-based access; Vercel Pro when function/computation demand justifies it.

13. Risks & mitigations

Risk	Mitigation (current)
LLM provider quota/price shifts	6-provider waterfall with free-tier floor; per-call pinning; Budget tier default
Vercel 12-function cap blocks growth	<code>?action=</code> routers + <code>_shared/</code> (new features $\approx$ 0 functions); CI gate counts functions on every PR; Pro upgrade is a decision, not a rewrite
Competitor capture blocked (anti-bot, inbox limits)	Multiple paths: IMAP poll, Cloudflare email webhook, off-Vercel Playwright workers; stagger/backoff; store-for-internal-benchmarking policy
Brand drift as generators multiply	Single brand kernel consumed by every generator; palette check + copy scrub at generation time; CI palette lock
Autonomy sends something wrong	Nothing sends today (Phase-2 gate); every Brain entry <code>needs_human_verification</code> ; approvals win over daily sync; weekly human recalibration is mandatory
Google-key leakage	Eliminated: WIF keyless auth; remaining secrets server-side only
Single-maintainer bus factor	This PRD; <code>docs/</code> runbooks (GO_LIVE, SESSION-HANDOFF, ARCHITECTURE); CLAUDE.md project memory; slash-command playbooks
iOS distribution friction (signing)	Unsigned IPA + sideload path now; store pipeline in Phase 2

14. Open questions

1. When does Phase 2 unblock — who owns the Klaviyo key and the send-approval policy?
 2. Auto-approve thresholds: what confidence × spend cap is leadership comfortable delegating?
 3. Store presence: publish the Android app publicly or distribute internally (APK link) only?
 4. CI legal posture at 50+ sources: confirm scope with counsel before scaling capture.
 5. Vercel Pro trigger: which concrete limit (function count, cron frequency, execution time) pulls the trigger?
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Appendix A — Route map

Route	Page/behavior
/	Home portfolio hub
/analytics	Data Analysis dashboard
/plan	30-day Marketing Calendar
/studio · /app · /mailer	Mailer Studio
/brain · /smart-brain	Smart Brain console
/chaigpt · /chai · /ask	ChaiGPT
/competitor · /discover	Competitor Benchmarking
/kb · /knowledge-base	Knowledge Base
/ads	Ad Campaigns
/landing · /landing-pages	Landing Pages
/assets	Created Assets hub
/cohorts	Cohort Definitions
/agent	Vahdam Agent (buyer-facing)
/lp/:id (+ `/lp/cortisol-v1` v2 , /lp/agent , /lp/best`)	
/deck	Original pitch deck · /prd-deck → this PRD's presentation deck
/api/health , /api/public-config	Health/config

Appendix B — API action surface (abridged)

- `api/brain.js`: ~30 actions — ops (`status` , `config`), KB, analysis (`analyze` , `cohorts` , `scores`), competitor (`benchmarks`), calendar (`calendar-generate` , `festivals` , `mvt`), generation (`generate` , `assets` , `campaigns`), review (`review` , `decide` , `recalibrate`), agents (`agent-chat` , `team-chat` , `jarvis`), `agentic-run` , `calendar-scenarios` , `brand-chat` , `brand-tools` , `klaviyo` , `tts` , `cron` .
- `api/calendar.js`: `generate` , `trigger-mailer` , `lp` , `smart-brain-{plan, sync-daily, preview, approve, reject, activate-scenario, run-daily, generate-slot, feedback, weekly-recalibration, ...}` .
- `api/competitor.js`: `list` , `html` , `poll` , `sync` , `seed` , `ingest` , `adlibrary` , `discover` , `ci-*` (collect/brands/ads/emails/landing/offers/enrich/funnel/daily).
- `api/kb.js`: `ingest` , `list` , `top-emails` , `brands` , `classify-emails` .
- `api/ai/*`: `generate` (concepts/brief/mailer_full), `image` (photo/design/ad), `pipeline/{strategy, variant, images, html, score}` .

Appendix C — Environment variables

Text: `OPENAI_API_KEY` (+ `_2` , `_3`), `ANTHROPIC_API_KEY` , `GEMINI_API_KEY` , `XAI_API_KEY` , `GROQ_API_KEY` , `CEREBRAS_API_KEY` . Storage: `SUPABASE_URL` , `SUPABASE_ANON_KEY` (+ service-role server-side). Lifecycle: `KLAVIYO_API_KEY` (+ `KLAVIYO_PUBLIC_KEY` , `KLAVIYO_REVISION`). Voice: `ELEVENLABS_API_KEY` . Google/Sheets: `GCP_WORKLOAD_IDENTITY_PROVIDER` , `GCP_SERVICE_ACCOUNT_EMAIL` , `GOOGLE_SHEET_ID` , `GOOGLE_SHEET_TAB` (legacy `GOOGLE_SERVICE_ACCOUNT_*`). Protection: `CRON_SECRET` , `INGEST_TOKEN` . Vercel-auto: `VERCEL` , `VERCEL_ENV` , `VERCEL_URL` , `VERCEL_OIDC_TOKEN` .

Appendix D — Screenshots

Captured 2026-07-02 from the running app (local preview of this repo):

!Home	!Analytics
!Calendar	!Mailer Studio
!Smart Brain	!ChaiGPT
!Competitor	!Knowledge Base
!Ad Campaigns	!Landing Pages